Benjamin Liu

Senior Software Engineer

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Summary

Senior Software Engineer with 10+ years of experience building scalable SaaS platforms across e-commerce, fintech, security, healthcare, and geospatial domains. I specialize in cloud-native architecture, distributed systems, and modern React/TypeScript/Node.js ecosystems, delivering secure identity platforms (SSO, OAuth, JWT, RBAC) and high-concurrency APIs optimized through observability (logging, monitoring, tracing). I lead cross-functional teams to deliver performant, reliable systems that drive measurable business impact.

Work Experience

PrimeiTek — Northridge, CA

Senior Software Engineer — Feb 2020 – Jan 2026

- Led a team of 6 engineers delivering large-scale customer-facing platforms across eCommerce/Subscription, fintech payments, and HIPAA-aligned healthcare workflows, owning architecture, performance, and end-to-end delivery from discovery to production.
- Architected and delivered end-to-end full stack solutions using Next.js, React, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, and Redis, scaling platforms to 5M+ daily operations with high availability and sub-second API response targets.
- Led system design for microservices and platform foundations: API gateway patterns, caching strategies, connection pooling, and horizontal scaling across AWS/Azure, improving API latency by 32% through profiling, query optimization, and cache tuning.
- Owned large eCommerce and subscription initiatives (catalog → checkout → post-purchase), integrating Stripe and PayPal payment flows (tokenization, webhooks, retries, idempotency), plus ACH fund movement patterns aligned with PCI compliance and secure authorization standards.
- Delivered HIPAA-aligned healthcare capabilities by implementing secure access patterns, audit-friendly workflows, and privacy-conscious architecture across internal tools and patient-facing experiences, collaborating closely with compliance-minded stakeholders.
- Drove growth outcomes by improving acquisition funnel and onboarding journeys through analytics-driven iteration, performance monitoring, and experimentation, increasing onboarding conversion by 18% across key touchpoints.
- Built and maintained design systems and reusable component libraries (React/TypeScript), translating Figma into accessible, production-ready UI with measurable improvements to frontend performance and maintainability.
- Designed distributed backend workflows including scheduling, queuing, rate limiting, and serverless task processing to support high-volume JSON pipelines and workflow orchestration with strong retry/timeout/error-handling standards.
- Implemented enterprise-grade Identity & Access Management: JWT, OAuth, SSO integrations, RBAC, and scalable permissions modeling; delivered team management and role-based authorization patterns for multi-tenant environments.
- Improved reliability and operational readiness by implementing structured logging, monitoring, tracing, and automated alerting, reducing production incidents by 27% and shortening MTTR via better runbooks and dashboards.
- Established coding standards and clean code practices; ran structured code reviews, pair programming, and mentorship plans, elevating test quality with test automation, refactoring strategies, and consistent CI/CD enforcement.
- Partnered cross-functionally with Product, Design, and customer-facing teams to scope requirements, prototype high-risk components, and deliver a major platform release to completion with clear milestones and predictable delivery.

Key Technologies: Next.js, React, TypeScript, Node.js, Express.js, RESTful APIs, GraphQL, Microservices, Distributed Systems, AWS, Azure, Docker, Kubernetes, CI/CD, Redis, MongoDB, PostgreSQL, Serverless Architectures, Workflow Orchestration, Observability (Logging, Monitoring, Tracing), Identity & Access Management (SSO, OAuth, JWT, RBAC), Stripe, PayPal, ACH, PCI Compliance, Design Systems, Figma, Technical Leadership, Mentoring

DroneDeploy San Francisco, CA
MERN Developer and Software Engineer Oct 2016 - Feb 2020

- Contributed as a MERN Developer and Software Engineer to the full stack delivery of cloud-based reality capture, drone mapping, and geospatial analytics software used by customers in construction, agriculture, energy, and inspection workflows.
- Developed scalable frontend features in React to support interactive viewing, annotation, and collaboration on geospatial maps and 3D site models, improving usability for inspection, progress tracking, and project documentation.
- Built and maintained backend services with Node.js and Express, implementing RESTful APIs for map processing, imagery uploads, and user-centric workflows across survey, construction, and monitoring use cases.
- Worked extensively within the MERN stack (MongoDB, Express, React, Node.js), contributing to data modeling, schema design, and query optimization for large-scale geospatial datasets and high-resolution imagery storage.
- Supported cloud processing workflows that transformed drone-captured aerial imagery into orthomosaics, digital elevation models, and analytics-ready geospatial outputs used for earthwork measurement, crop analysis, and site documentation.
- Implemented reverse proxy and traffic routing configurations using NGINX to support service reliability and distributed cloud environments.
- Developed secure authentication and OAuth-based identity flows for enterprise customers, ensuring safe access to mission data and map assets.
- Collaborated with DevOps to optimize AWS EC2, Auto Scaling Groups, and load balancing strategies for scalable processing workloads.
- Communicated technical decisions and implementation trade-offs with product, design, and data science teams to align frontend performance, backend efficiency, and user experience goals.

Key Technologies: React, Node.js, RESTful APIs, AWS (EC2, Auto Scaling, Load Balancing), NGINX, OAuth, Database Architecture & Query Optimization, Distributed Systems, Cloud Deployment, Geospatial Mapping & Analytics

WLCM App Studio San Francisco, CA
Frontend UI/UX Developer Oct 2013 - Oct 2016

- Contributed to the development of education platforms and analytics tools for early-stage SaaS products, focusing on intuitive user interfaces and seamless learning workflows from MVP to production.
- Implemented component-driven frontend development using modular, reusable UI patterns with JavaScript (ES5/ES6), HTML5, and CSS3, delivering responsive student, teacher, and admin experiences with strong emphasis on usability and accessibility.
- Collaborated closely with product and design stakeholders to translate wireframes and UX concepts into production-ready frontend implementations, ensuring consistent UI behavior across dashboards and reporting modules.
- Built interactive data visualization modules using D3.js for educational reporting and performance dashboards, enhancing data clarity and user engagement.
- Integrated frontend applications with RESTful APIs built using Node.js, Express.js, and PostgreSQL, working alongside backend engineers to ensure reliable data flow and reporting pipelines.
- Applied Test-Driven Development (TDD) practices using Mocha/Chai to improve frontend reliability and reduce regression risk during feature enhancements.
- Participated in CI/CD processes with Jenkins and supported AWS deployments to maintain stable releases and consistent user experiences across environments.
- Contributed within Agile/Scrum teams, engaging in sprint planning, backlog refinement, and iterative UI improvements aligned with product and user feedback.

Key Technologies: Angular, Node.js, Express.js, PostgreSQL, D3.js, RESTful APIs, TDD (Mocha/Chai), Jenkins CI/CD, AWS, Full Stack Development, Education Platforms

Education

University of California, Davis 2009 - 2013
Bachelor of Science (BS), Computer Science Engineering

Skills

Frontend:
React.js, Next.js, Vue.js, TypeScript, JavaScript, Redux, Context API, Zustand, HTML5, CSS3, Responsive Design, SPA Architecture, Progressive Web Apps (PWA), Frontend Development, Vite, Tailwind CSS, Sass/SCSS, shadcn/ui, Modern React Patterns, Design Systems, Component Libraries, Storybook, Frontend Performance Optimization, Clean Code Practices, Figma Collaboration, UI/UX Systems, Visual Design Implementation, MUI (Material UI), Digital Products Development, Interaction Design, Product Design Principles, User Experience (UX), UX/UI, UX Design, React Native, Mobile Development
Backend:
Node.js, Express.js, RESTful APIs, GraphQL, Backend Development, Workflow Orchestration, Distributed Task Processing, Scheduling & Queuing Systems, Rate Limiting, Serverless Architectures, Algorithms, Microservices Architecture, SOLID Principles, Data Structures, Design Patterns, Payment Systems Integration (Stripe, ACH), Fintech Platforms, SSO (Single Sign-On), Team Management Systems, Test Infrastructure, Linux/Unix Operating Systems, API Integration, OpenAPI/Swagger, Java, Python, Django, Go
Forms & Validation:
Formik, Zod, Schema Validation, Secure Input Handling
Databases & Data:
MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch, Firebase/Firestore, Supabase, Database Architecture, Query Optimization, JSON Processing, Healthcare Data Standards (HL7, X12, EDI), Large-Scale Data Pipelines, Data Modeling, Indexing, Performance Optimization
DevOps & Cloud:
AWS, Azure, GCP, Azure AI Services, Docker, Kubernetes, Infrastructure as Code (Terraform), Continuous Integration, Continuous Deployment, CI/CD, GitLab CI/CD, DevOps Practices, Serverless Deployment, Structured Logging, Observability, Tracing, Enterprise AI Deployment, Git
Testing & Quality:
Test-Driven Development (TDD), Jest, React Testing Library, ESLint, Prettier, Automated Testing, Code Quality Enforcement
Security:
JWT, OAuth, RBAC, Session Management, API Security, PCI Compliance, Secure Fund Flows, Payment Authorization Standards, SSO, Identity & Access Management (IAM), Permissions Modeling, AI Security, Prompt Injection Prevention, LLM Security Considerations
Architecture:
Full Stack Architecture, Distributed Systems, System Design, High-Concurrency Systems, Scalability Engineering, Identity Systems, Authorization at Scale, Refactoring Practices
Methodologies & Soft Skills:
Agile, Scrum, Agile Software Development, Code Review, Technical Mentorship, Coaching, Ownership, Team Collaboration, Innovation, Strategic Thinking, Decision-Making, Adaptability, Self-Starter Mindset, Written & Verbal Communication, Feedback-Driven Development, Software Development Lifecycle (SDLC), Testing (Unit testing, Integration, Jest), Problem Solving, Advanced Analytical and Problem-Solving Techniques, Technical Research, AI Solution Architecture, Continuous Improvement, Ethics, Attention to Detail, Code Quality
AI & Artificial Intelligence:
AI-Assisted Development (GitHub Copilot, ChatGPT, Cursor), OpenAI API Integration, Claude API Integration, LLM Integration, Prompt Engineering, LangChain, LLM Orchestration Frameworks, AI Development & Platform Architecture, AI Agents, AI-Augmented Engineering Workflows
Blockchain & Web3:
Ethereum, Web3.js, Ethers.js, Smart Contract Integration, Cryptocurrency, DeFi, DEX Architecture